SE 321/SE 305
SOFTWARE SPECIFICATION AND DESIGN

Activity Diagrams and Use Case Diagrams

WEEK #5

SECOND ASSIGNMENT

- Please make an appointment with your clients. Conduct your interviews.
- Write down your findings and analyze them.
- Extract your functional and non-functional requirements.
- Prepare and write your requirement documents.

SECOND ASSIGNMENT (cont'd)

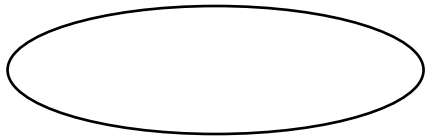
- Prepare Meeting Report #1, change its name to 'MeetingReport1_ProjectName_SectionNumber', convert to PDF, and send it to me.

- Write your requirements to Meeting Report #1 attachment section. DO NOT SEND ME requirement specification forms please!!!

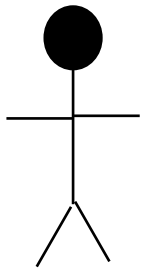
- Due date: 27th of October @23:59

- Also print a copy of it and bring it to the LAB hours next week.

Elements in Use Case Diagrams



- Use Case



- Actor

- Boundary
- Connection

- Include relationship

<<include>>

Note: different names used in different software

- Extend relationship

<<extend>>

Use Case Diagram

- A diagram representing system's behavior—use cases and actors.
- Provides a global look of a system – it's basic functionality (use cases) and environment (actors).
- Useful for early structuring of requirements; iterative revisions.
- May be understood by users on the intuitive level.

Extend Relationship Between Use Cases

- Extend relationship –

  linking an *optional* use case to a standard use case.

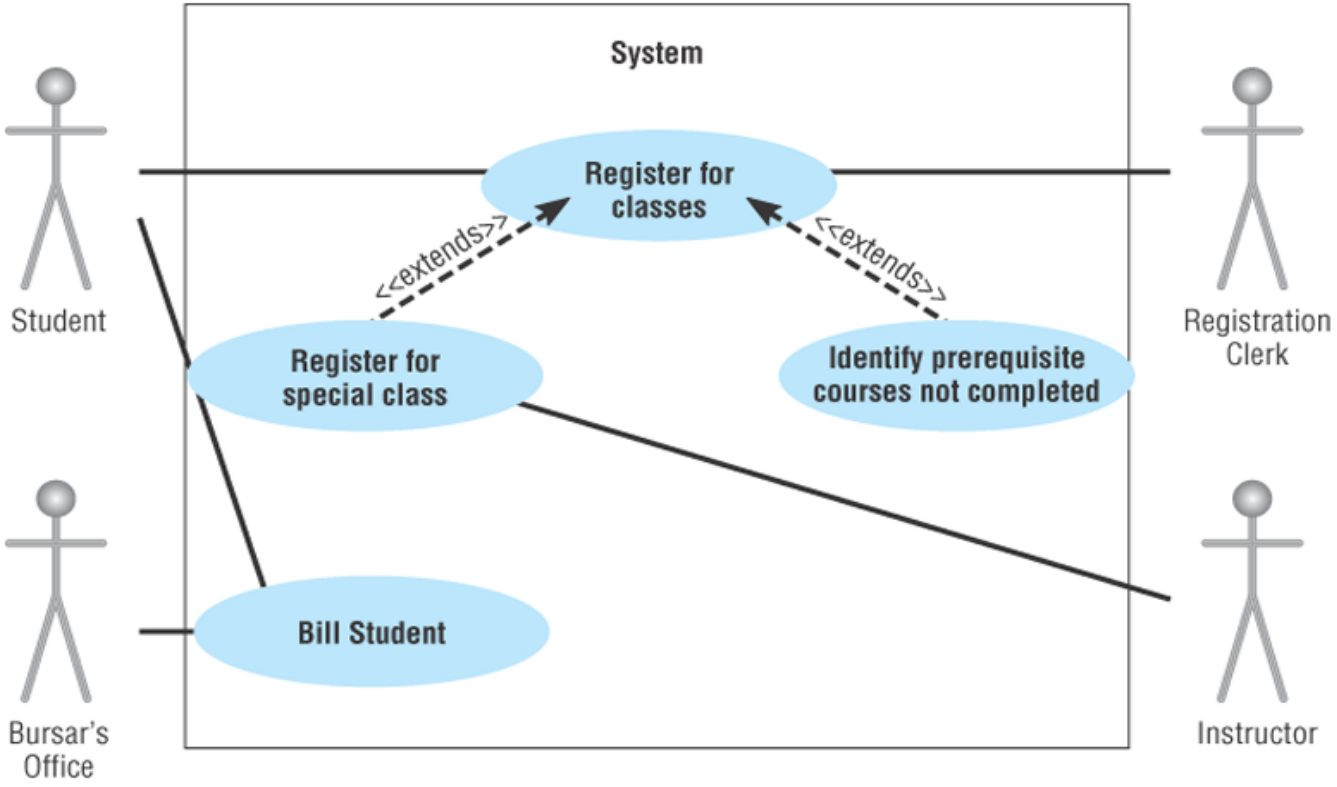
Example: *Register Course* (standard use case) may have *Register for Special Class* (extend use case).

- Standard use case can execute without the extend case.

→ loose coupling

Extend Relationship Between Use Cases (cont'd)

Figure 6.4 A Use Case Diagram for a University Registration System Represented with Microsoft's Visio



Careful: Arrows between Use Cases are NOT data flows!

Include Relationship Between Use Cases

- Include relationship –

  a standard case linked to an *mandatory* use case.

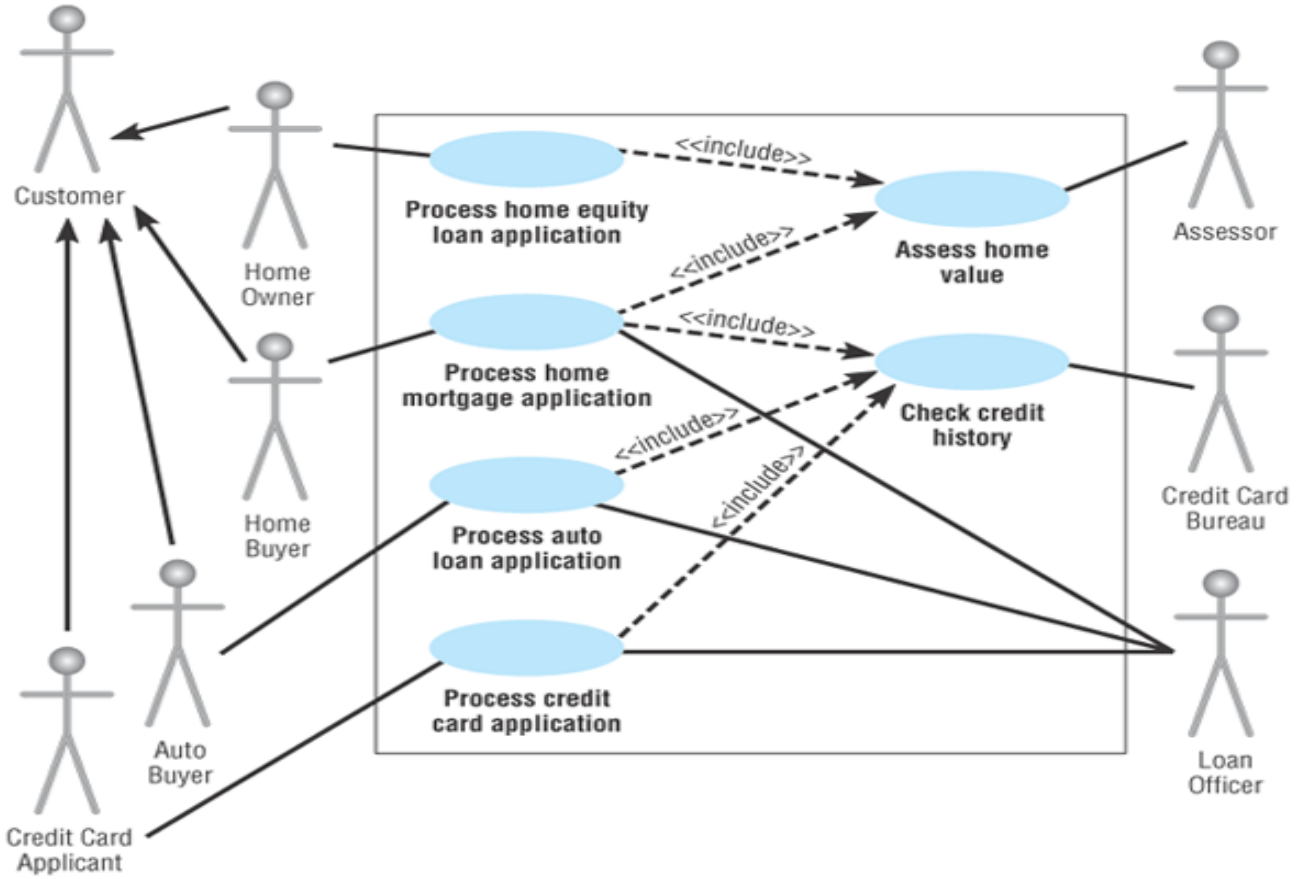
  Example: to *Authorize Car Loan* (standard use case), a clerk must run *Check Client's Credit History* (include use case).

  → tight coupling

- Standard use case can NOT execute without the include case.

Extend Relationship Between Use Cases (cont'd)

Figure 6.6
Examples of Generalized Use Cases and Include Relationships



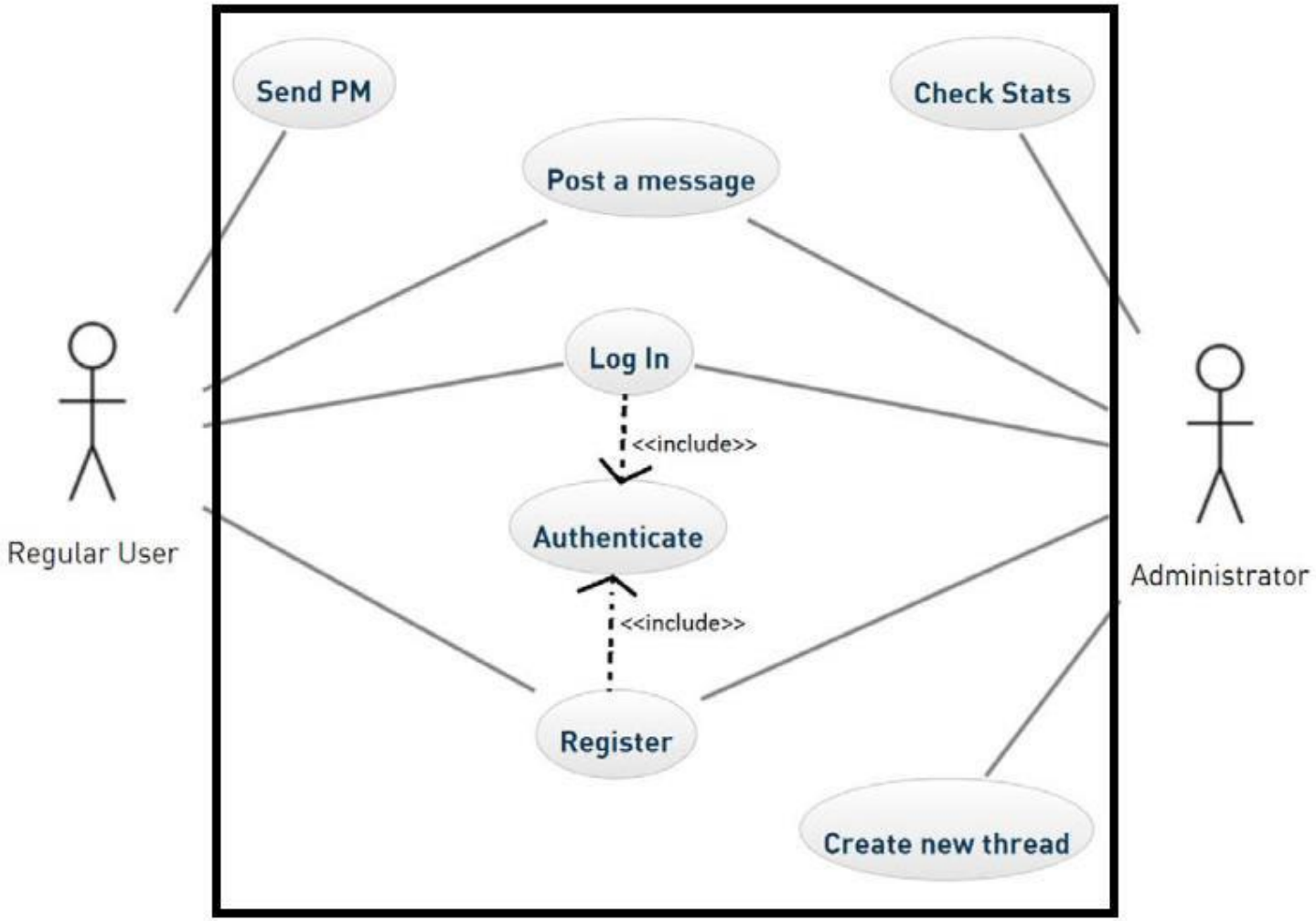
Careful: Arrows between Use Cases are NOT data flows!

Example #1: Online Forum

Imagine that you are analyzing requirements for an online forum system. Forums can get very complex, but imagine that we have only two kinds of users that interact with our system with different responsibilities: Regular Users and Administrators. Both can log in to the system, and part of logging in is an internal authentication process. Both can also register with the system, which also uses internal authentication. After logging in, everybody can post new messages to the board, however only Administrators can check statistics and create new threads. Regular users on the hand can send private messages to other users, while administrators do not have this ability.

Draw a Use Case diagram that contains Actors, Use Cases and their relationship from the scenario.

Example #1: Online Forum



Activity Diagrams

Activity Diagrams

- An activity diagram is a type of flowchart that is part of the UML (Unified Modeling Language) standard.
- Its purpose is to enable analysts to present a concrete, easy-to-follow visual of the workflow of a business use case.
- A diagrams may be used to describe an entire system, a use case, or an activity within the use case.
- They also enable the simple presentation of multiple conditions and choices within a workflow.

Elements in Activity Diagrams

- Initial nodes, which indicate the start of the workflow (also known as initial state). This is represented by a blackened circle.
- End nodes, which indicate the end of the workflow (also known as the final state). These are represented by a blackened circle within a circle.
- Control flows, which indicate the direction of the workflow. This is represented by an arrow.
- Activities, which indicate steps or stages in the workflow. These are represented by ovals.

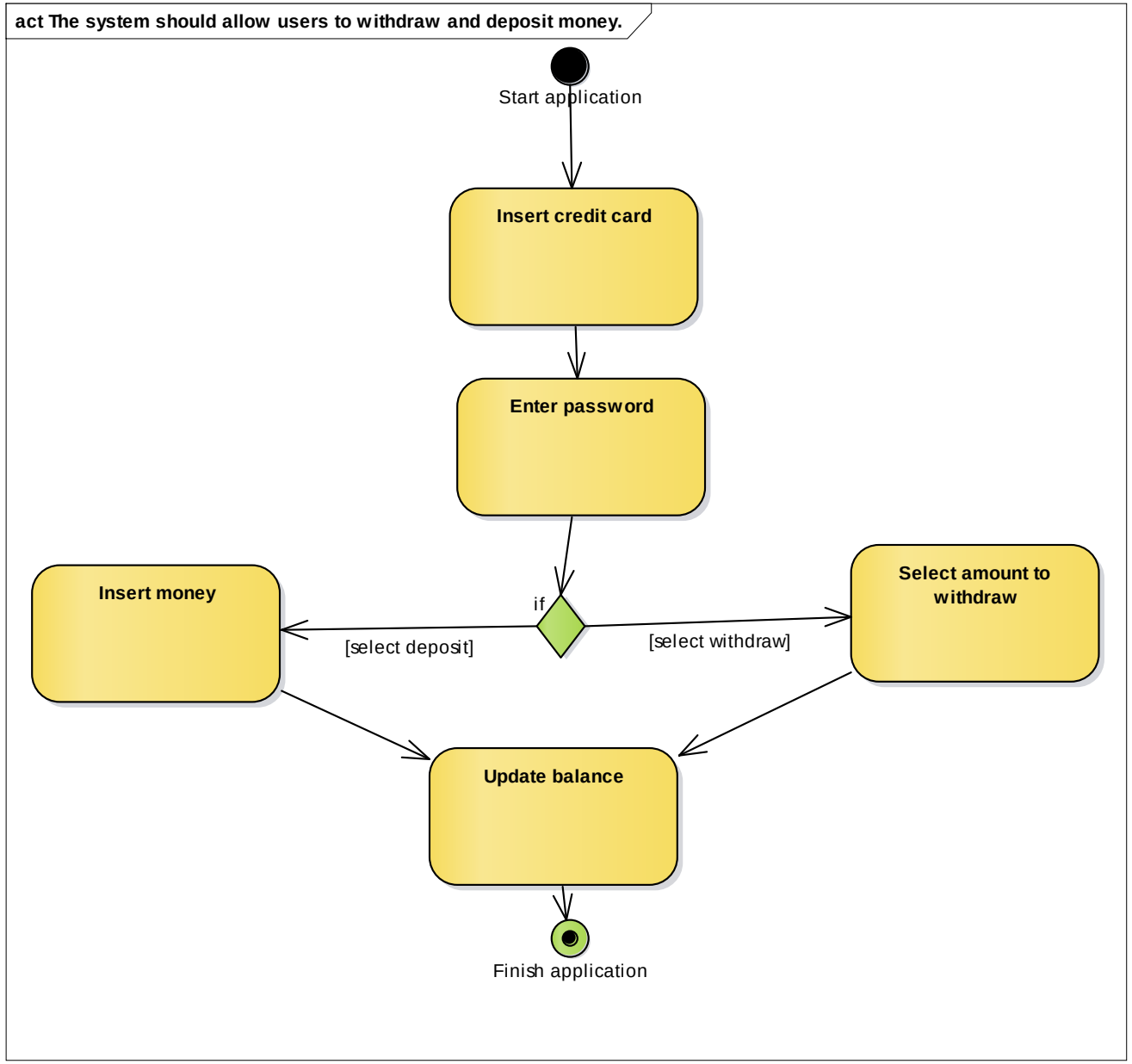
Elements in Activity Diagrams (cont'd)

- Decisions, which indicate a choice between multiple conditions. These are represented by diamond shapes.

- Guard conditions, which indicate a condition required for the workflow to proceed, and which normally (but not always) follows a decision. These are represented by brackets.

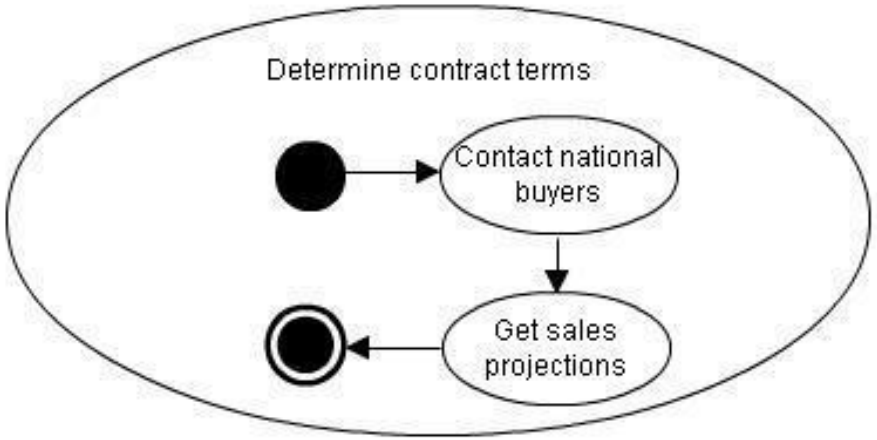
- A fork and join, which indicates two or more possible conditions that may occur in the same process. Both the initial split (or fork) and the latter join are represented by the same symbol, a horizontal bar.

Example #1


act The system should allow users to withdraw and deposit money.
Start application
Insert credit card
Enter password
if
[select deposit]
[select withdraw]
Insert money
Select amount to withdraw
Update balance
Finish application

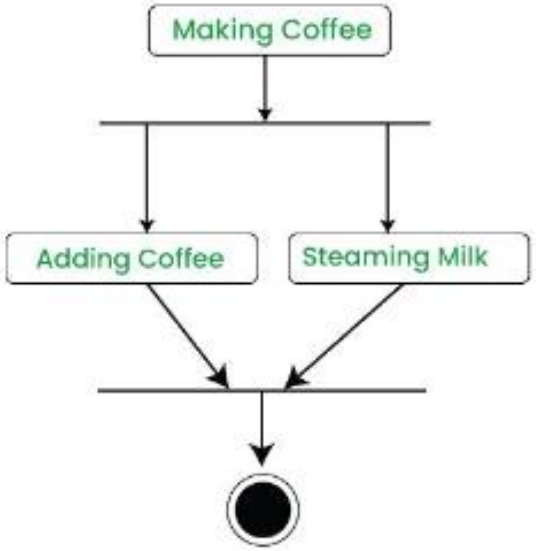
Example #2

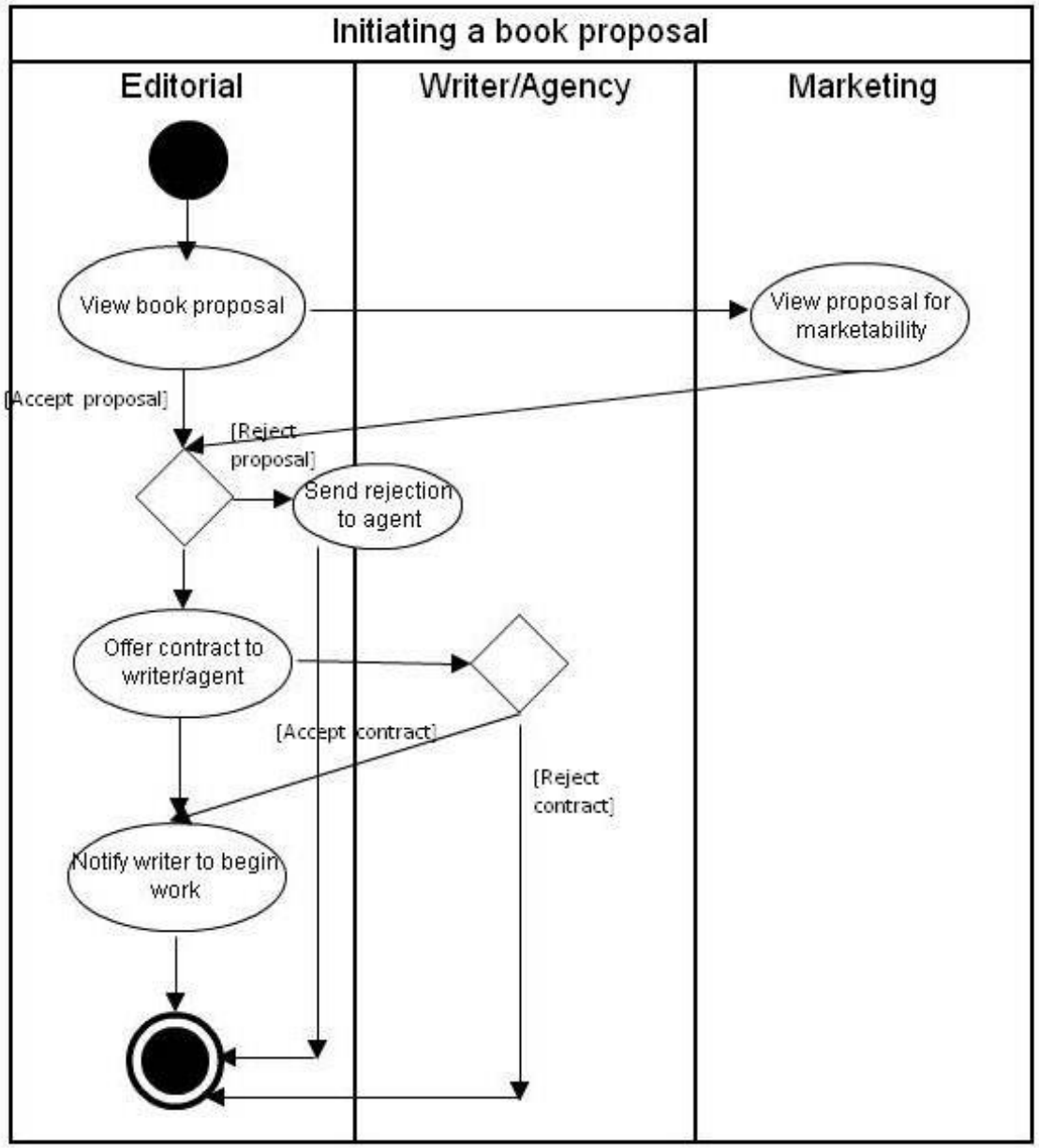
- Other possible (but not always necessary) elements of an activity diagram include nested activities (basically a small activity diagram within an activity diagram).



Example #3

A Diagram using Join Notation



Example #4

Another variation of an activity diagram is a diagram with swimlanes, which are partitions that indicate who does what—displaying an actor or entity in each lane.